1.REFRACTION OF LIGHT

Water, glass, air, diamond, etc. are transparent to light, meaning they allow light to pass through them. However, the speed of light in each of these materials is different. See the table given below.

Medium	Speed of light
Air/Vacuum	$3x10^8$ m/s
Water	$2.25x10^8$ m/s
Glass	$2x10^8$ m/s
Diamond	$1.25x10^8$ m/s

Optical density

The ability of a medium to influence the speed of light through it is its optical density.

The speed of light decreases as optical density increases.

The increasing order of optical densities of the media is given below.

Air<Water<Glass<Diamond

It is noted that a substance with high material density may not have high optical density. For example, the density of water is more than that of kerosene. However, the optical density of water is less than that of kerosene.

Refraction

Typically, light travels in a straight line. However, under certain conditions, it deviates from this path. See the figure.1

A light ray enters from air to water obliquely. The ray undergoes deviation at the surface of separation.

It is noted that it doesn't undergo deviation if the ray enters normally from one medium to another.

(See fig.2)

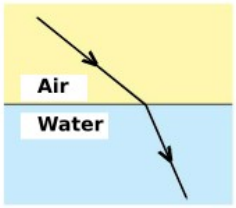


Fig.1

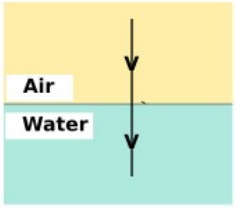


Fig.2

When a ray of light enters obliquely from one medium to another with different optical densities, it undergoes deviation at the surface of separation. This phenomenon is called refraction.

Cause of refraction

Refraction of light is caused by the change in its speed when it passes from one medium to another. This change in speed occurs due to the change in the optical density of the medium.

Refractive index: Refractive index of a medium is the ratio of the speed of light in vacuum to the speed of light in the medium.

Refractive index, n = c/v, where c – speed of light in vacuum & v – speed of light in the medium.

Layman's Science Magazine

Sample Questions.

?1. Find refractive index of water, air and diamond using the table given.

Medium	**Speed of light**
Air/Vacuum	$3x10^8$ m/s
Water	$2.25x10^8$ m/s
Glass	$2x10^8$ m/s
Diamond	$1.25x10^8$ m/s

Ans.i. Refractive index of water = speed in vacuum/speed in the water = $3x10^8/2.25x10^8 = 1.3$

ii. Refractive index of air = speed in vacuum/speed in air = $3x10^8/3x10^8 = 1$

iii. Refractive index of diamond = $3x10^8/1.25x10^8 = 2.4$

?2. Refractive index of a substance is 1.5 Find the speed of light through that substance.

Ans. We have, n = c/v

Then v = c/n = $3x10^8$ /1.5 = $2x10^8$ m/s

?3. See the table given below.

medium	**Speed of light**
Crown glass	$1.97x10^8$ m/s
Glycerine	$2.04x10^8$ m/s
Sunflower oil	$2.04x10^8$ m/s
Water	$2.25x10^8$ m/s
Flint glass	$1.85x10^8$ m/s

a. Which medium has the highest optical density?

b. Which has the lowest optical density?

c. Is there refraction when light passes obliquely from glycerine to sunflower oil? Justify your answer.

Ans.a. Flint glass has the highest optical density because light travels at the lowest speed through this medium. b.water.

c. No, there will not be refraction because there is no difference in optical density between glycerine and sunflower oil.

Relation between speed of light and refractive index of a medium.

Speed of light decreases with refractive index.

That is, light travels faster through the medium of lower refractive index.

Diamond has the highest refractive index among materials, and hence the speed of light through diamond is the lowest.

?4 How is optical density related to refractive index?

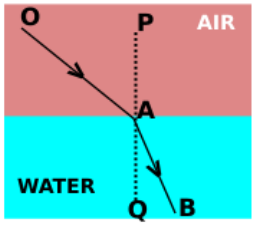
Ans. A substance with higher optical density will have a higher refractive index.

Layman's Science Magazine

Terms related to refraction

i. Incident ray: An incident ray is the ray that strikes the plane separating two mediums. In the given figure, OA is the incident ray

ii. Refracted ray: The ray that passes through the second medium after refraction is the refracted ray. AB is the refracted ray.



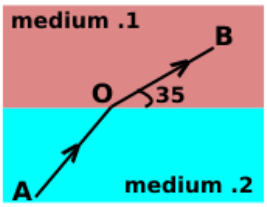
iii. Angle of incidence(i): The angle between incident ray and the normal drawn at the point of incidence is angle of incidence.

In the figure, <OAP is the angle of incidence

iv. Angle of refraction(r): It is the angle between refracted ray and the normal.

<QAB is the angle of refraction.

?5.See the figure.



a. Identify incident ray and refracted ray.

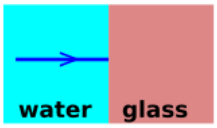
b. What is the angle of refraction?

Ans.a. Incident ray: AO Refracted ray: OB

b. Angle of refraction is the angle between refracted ray and normal.

Then r = 90-35 = 55°

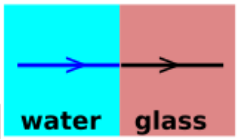
?6.A light ray is incident at a glass slab from water as shown.



a. Complete the path of the ray.

b. What is the angle of incidence?

Ans.a.



b. Angle of incidence is zero.

Direction of bending of refracted ray

When light enters from a rarer medium to a denser medium (For example: Air to water), the refracted ray bends towards the normal. See Fig.A

When light enters from a denser medium to a rarer medium (For example: glass to air), the refracted ray bends away from the normal. See Fig.B

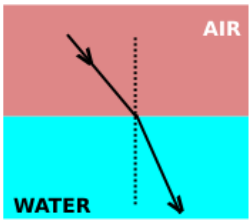


Fig.A

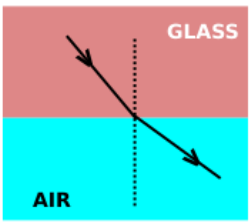


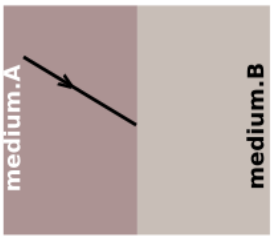
Fig.B

Note:i. When light travels from an optically rarer medium to a denser medium, angle of refraction will be less than the angle of incidence.

ii. When light travels from an optically denser medium to a rarer medium, angle of refraction will be greater than the angle of incidence.

Layman Science Magazine

?7.A light ray is incident from a medium A to another transparent medium B. The optical density of A is greater than that of B.

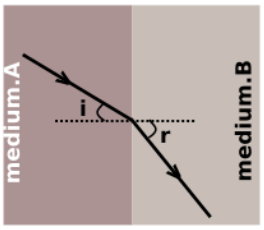


a. Complete the path of light ray and mark the angle of incidence(i) and the angle of refraction(r).

b.Between the two mediums, in which medium will the speed of light be greater? Justify your answer.

c. Which will have greater refractive index?

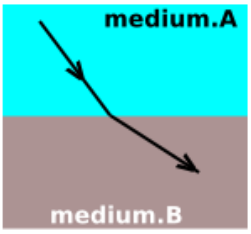
Ans.a.



b. Speed will be greater in the medium B. Because speed is greater in the medium of less optical density.

c. Medium A will have greater refractive index.

?8. A ray of light passes from a medium A to another medium.B



a. Identify the phenomenon.

b.The refracted ray bends .....

(towards the normal/away from the normal)

c. Identify the optically denser medium. Justify your answer.

d. In which medium will the speed of light be faster?

Ans.a. Refraction. b.away from the normal

c. medium A. It is when light travels from a denser medium to a rarer medium that the refracted ray bends away from the normal.

d. Since B is the rarer medium, speed of light will be faster in B.

?9. When light travels from glass to diamond, the angle of refraction will be ..... angle of incidence. (less than /more than) **Ans**.a. less than .

?10. “As the angle of incidence increases, the angle of refraction increases."

This is true:

a. only when light travels from a denser medium to a rarer medium.

b. only when light travels from a rarer medium to a denser medium.

c. in both cases.

Ans. in both cases.

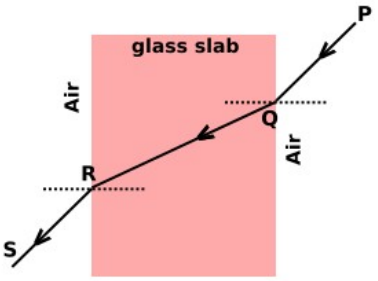
Layman's Science Maga

Refraction in a glass slab

It is given the path of light ray incident obliquely at a glass slab.

Here the light undergoes two refractions.

At first, light ray enters from air to glass(from rarer medium to denser medium). Then the refracted ray QR bends towards the normal. At R, light enters from glass to air and hence the refracted ray RS bends away from the normal.



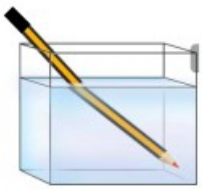
Daily experiences due to refraction of light.

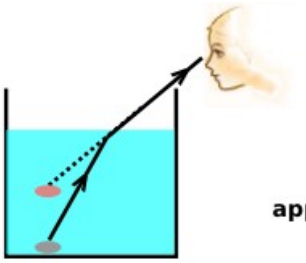
1.A straw or a pencil appearing bend when it is partially dipped in water. (See Fig.a)

2. An object lying under water or any transparent liquid appears to be raised. (See Fig.b)

3. The bottom of a pond appears to be elevated when viewed from distance.

4.People who engage in bowfishing,aim at a point slightly below the perceived position of the fish. (Fig.c)


Fig.a


Fig.b

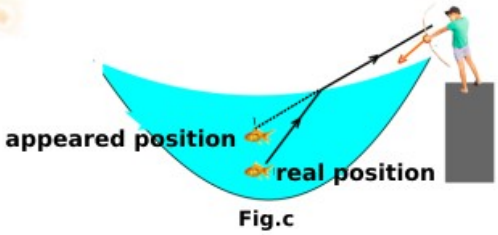


Fig.c

Atmospheric refractions

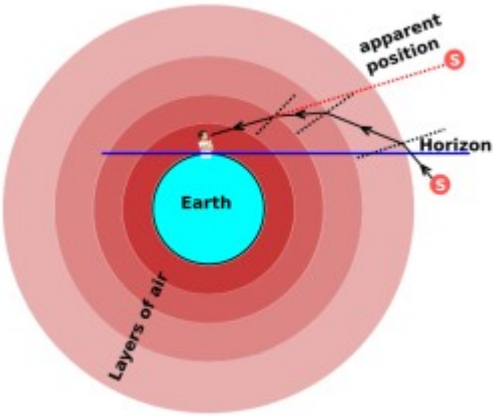
1. **The twinkling of stars**

Atmospheric refraction causes the stars to appear to twinkle. The light coming from the stars reaches our eyes by traversing through several layers of the atmosphere with varying optical densities. During this journey, the light rays undergo repeated and irregular refractions, and hence the stars cannot be seen continuously at the same position. Thus, the stars appear to twinkle.

2. **Sun is visible a little before actual sun rise.**

Due to atmospheric refraction, the Sun can be seen a few minutes before it reaches the eastern horizon in the morning. This phenomenon is depicted in the figure. When sunlight passes through the different layers of atmosphere of varying optical density,it undergoes slight refractions, causing the position of the Sun to appear higher than its actual position. As a result, the Sun becomes visible before it actually rises.

Similarly, the Sun is visible for a few minutes even after it has set in the evening.

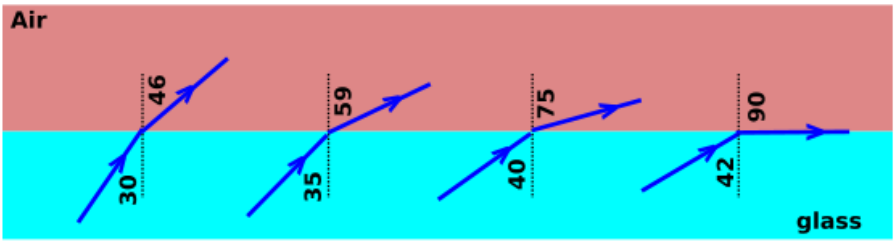


Layman's Science Magazine

Critical Angle(C)

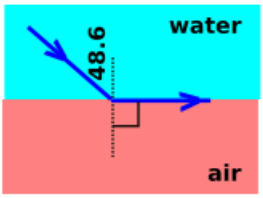
Observe the refraction of light when it travels from an optically denser medium (glass) to a less dense medium (air) at gradually increasing angles of incidence.

As the angle incidence increases, the refracted ray bend more and more away from the normal and hence gradually increases the angle of refraction.



When the angle of incidence becomes a particular value, the refracted ray just grazes along the surface of water. That is, the angle of refraction becomes 90°

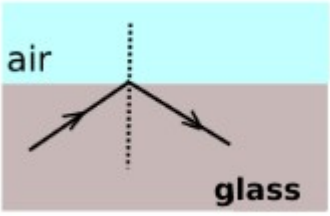
When light travels from an optically denser medium to a rarer medium, the angle of incidence at which the angle of refraction becomes 90° is the critical angle.



Critical angle of glass – air pair is 42° and that of water – air pair is 48.6°

Total internal reflection.

When light travels from an optically denser medium to a rarer medium, if the angle of incidence is greater than the critical angle, it is completely reflected back into the same medium without undergoing refraction. This is called total internal reflection.



Note: Two conditions are to be satisfied for total internal reflection.

i. Light should incident from an optically denser medium to a rarer medium.

ii.Angle of incidence should be greater than the critical angle of that pair of mediums.

?11.Complete the following using the terms given in the bracket.

a. Total internal reflection occurs when the angle of incidence is .....

(equal to critical angle/greater than critical angle/less than critical angle)

b. If total internal reflection occurs when the angle of incidence is 30°, which of the following is likely to be the critical angle? (24°/30°/42°)

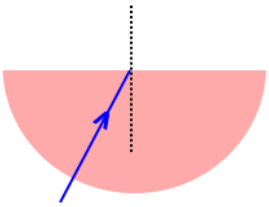
c. When total internal reflection occurs in the water – glass pair, light will be reflected back

(from water to water/ glass to glass/ either glass to glass or water to water)

Ans.a. greater than critical angle. b. 24° d. glass to glass.

Layman's Science Magazine

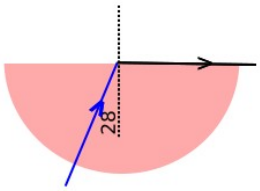
?12.Light from a laser torch is incident on a semicircular transparent object with a critical angle of 28°.

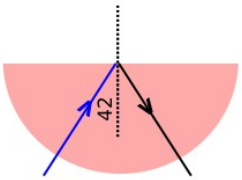


Complete the path of light if:

i. Angle of incidence is 28° ii. Angle of incidence is 42°

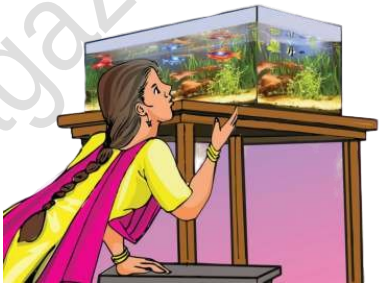
Ans.





Natural phenomena due to total internal reflection.

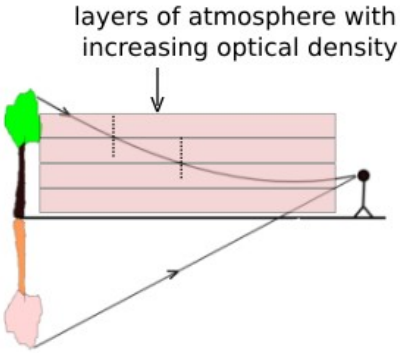
1.It is well-known that the bottom of the aquarium appears to be above the surface of the water (see the figure). This is due to the total internal reflection of light coming from the bottom of the aquarium at the water–air interface.



Here, the free surface of the water acts as a plane mirror when viewed from the bottom.

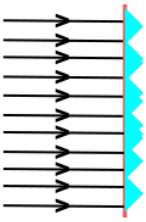
2. Mirage

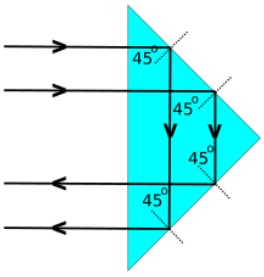
It is an optical illusion observed in hot deserts and on tarred roads, which gives the impression of a pool of water. On hot summer days, the air near the ground becomes much hotter than the air above. As a result, the optical density of air layers decreases with height. Light coming from tall objects such as trees or clouds bends more and more away from the normal and eventually undergoes total internal reflection. The inverted image of these objects appears on the road, giving a distant observer the illusion of water.



Applications of total internal reflection.

1. **Reflector**

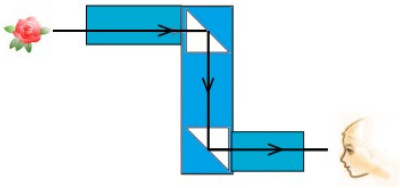




Total internal reflection is used in the reflectors of bicycles and the tail lamps of vehicles. A large number of small right-angled prisms are fixed in them. Since the critical angle of the substance used to make the prism is less than 45°, the light incident on the prisms is completely reflected back due to total internal reflection, as shown.

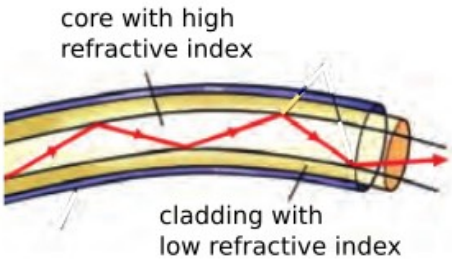
Layman's Science Magazine

2. **Periscope:**A periscope can be made using right-angled prisms, as shown, instead of plane mirrors.



3. Optical fibres.

Optical fibres are widely used in the field of telecommunication, utilizing total internal reflection of light. An optical fibre is a narrow strand with a core made of transparent material of high optical density, surrounded by cladding made of material with low optical density, as shown



Thousands of optical signals can be transmitted simultaneously to distant places at the speed of light without loss of intensity.

Optical fibres are also used in the medical field, such as in endoscopy.

Reflection from a plane mirror	**Total internal reflection**
Light is partially reflected.	Light is completely reflected.
Reflection occurs at any angle of incidence	Reflection occurs only when angle of incidence is greater than critical angle.
Intensity of reflected light decreases	There is no loss in the intensity of reflected light.

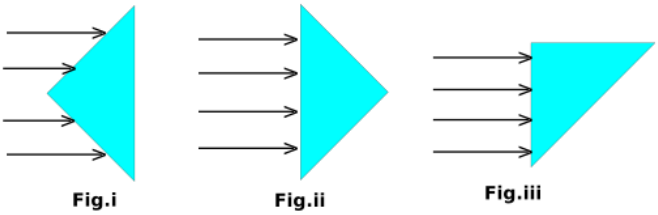
?13. Find the odd one from the following. How does it differ from others?

(mirage, twinkling of stars, ponds are seen as shallower, Sun can be seen a few minutes earlier at morning before it reaches horizon)

Ans.mirage. It is observed due to total internal reflection of light, while other optical illusions are typically caused by refraction.

?14.Which of the following is the proper arrangement of prisms in reflectors used on bicycles?

Ans. Fig.ii



?15. Find out the wrong pair from the following.

i. Optical fibre – total internal reflection ii. mirage – total internal reflection

iii. The letters on the paper under the glass slab appear raised.– refraction

iv. The bottom of the aquarium is seen on the surface of water - refraction

Ans. iv. The bottom of the aquarium is seen on the surface of water - refraction

Layman's Science Magazine

?16. Which of the following is the correct statement related to optical fibres?

a. The refractive index of the core is greater than that of the cladding.

b. The refractive index of core is less than that of the cladding.

c. The refractive index of core is same as that of the cladding.

Ans. a. The refractive index of the core is greater than that of the cladding.

?17. Which of the following cannot work as periscope?

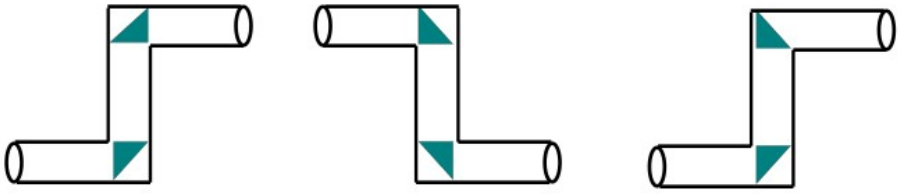


Fig.1 Fig.2 Fig.3

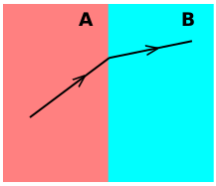
Ans. Fig.3

LET's ASSES

1. Ray diagram showing the path of light through mediums A & B is given.

a. In which medium will the speed of light be less – A or B?

b. Which will be optically denser medium? Justify your answer.

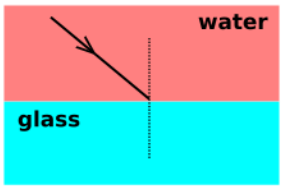


Ans.a. Medium B.

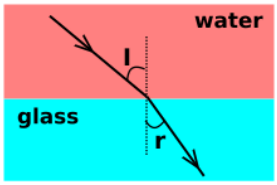
b. Medium B, because the refracted ray bends towards the normal.

It occurs when light travels from an optically rarer medium to a denser medium.

2. Complete the given diagram. Mark the angle of incidence angle of refraction.



Ans.



i - angle of incidence
r - angle of refraction

Light passes from medium X to Y. Here, the angle of refraction is greater than the angle of incidence.

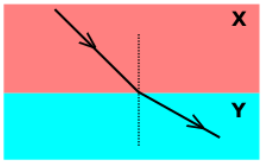
a. In which medium is the speed of light more? b. Which is the medium of larger refractive index?

c. Draw the path of light.

Ans.a. Y [Since r >i, light is travelling from optically denser medium to rarer medium]

b. X [Refractive index is greater for optically denser medium]

c.



Science Magazine

4. Refractive index of different mediums are given in the table.

Medium	Refractive index
Crown glass	1.52
Glycerine	1.47
Sunflower oil	1.47
Water	1.33
Flint Glass	1.62

a. In which medium does light travel with maximum speed?

b. Will a ray of light entering obliquely from glycerine to sunflower oil deviate? Explain the reason.

c. Light is transmitted from flint glass to each medium listed in the table given. If light is incident at an angle of 30°, which medium will have the largest angle of refraction? Why?

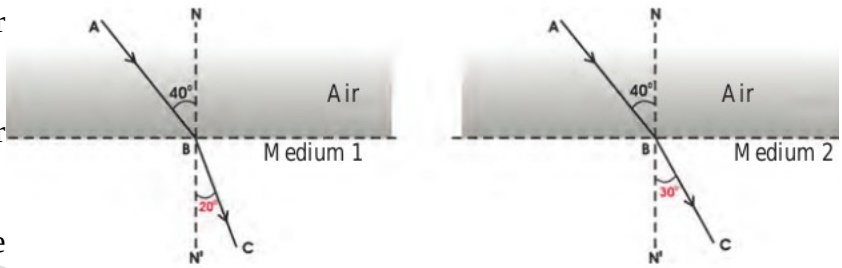
Ans.a. Water b. No. Because there is no difference in optical density.

c. water. Here, water is the medium having least refractive index. So refracted ray bends away more from the normal when light enters from flint glass to water.

5. Observe the figure. Ray of light incident on two mediums is depicted.

a. Which is the medium of higher optical density. Why?

b. Which is the medium of greater refractive index?



Ans.a. Medium.1 Because the refracted ray bends more towards the normal in this medium.

b. Medium.1. Refractive index of optically denser medium will have greater refractive index.

6. Observe the figures.

a. Which figure indicates total internal reflection? b. Which figures indicates refraction?

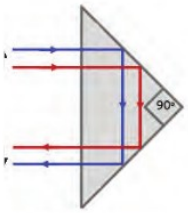


Fig. 1.29 (a)

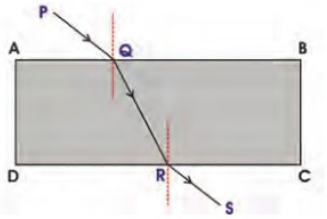


Fig. 1.29 (b)

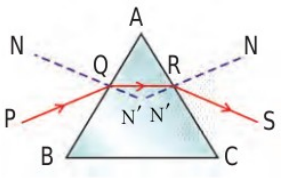


Fig. 1.29 (c)

Ans.a. Fig.1.29(a) b. Fig.1.29(b)& Fig.1.29(c)

7. The critical angle of glass is 42°. Choose the angle of incidence for which total internal reflection takes place? a. 40° b.49° c.38° d. 42° **Ans**.49°

[Total internal reflection occurs only when the angle of incidence exceeds the critical angle.

For remaining units - Contact: 9495676772

8. We can see small prisms in the reflectors used on motor cycles. Describe the benefits of using them.

Ans. Right-angled prisms used in reflectors reflect light very efficiently through total internal reflection. This improves the visibility of the motorcycle, even in low-light conditions, and hence reduces the possibility of accidents.

9. Observe the table.

Medium	Refractive index
Air	1.0003
Water	1.33
Kerosene	1.44
Turpentine oil	1.47
Crown glass	1.52
Diamond	2.42

a. Choose the medium in which light has the least speed.

b. The speed of light in air is $3x10^8$ m/s. What is the speed of light in kerosene?

c. When a ray of light enters obliquely from air to diamond, will the refracted ray deviate towards the normal, or away from the normal? Justify your answer.

Ans.a. Diamond. [Because of its greatest refractive index/optical density]

b. We have $n = c/v$

Then speed in kerosene, $v = c/n = 3x10^8/1.44 = 2.08x10^8$ m/s

c. It will deviate towards normal. Because light is travelling from an optically rarer medium to a denser medium.

10. The path of light through mediums A,B,C and D are given. Choose the correct figures.

(The optical density of the mediums are in the order A<B<C<D)

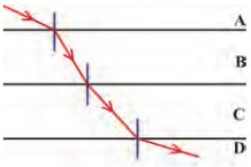


Fig. 1.30 (a)

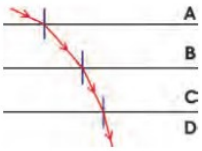


Fig. 1.30 (b)

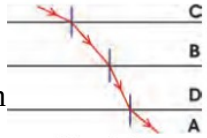


Fig. 1.30 (c)

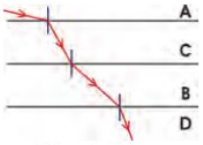


Fig. 1.30 (d)

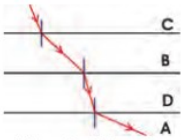


Fig. 1.30 (e)

Ans. Figures. 130(b),(d)&(e) are correct.

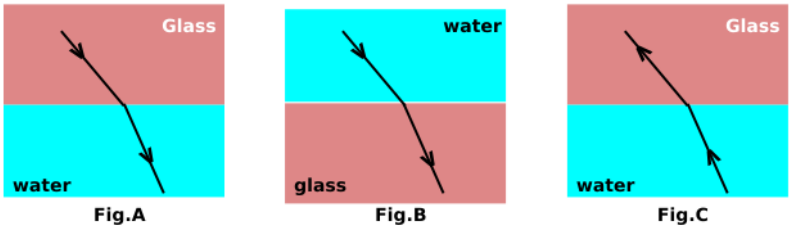
11. Speed of light in ethanol is lesser than that in methanol. Which medium has lower refractive index?

Ans. Methanol. Because speed of light is faster in the medium of lower refractive index.

Layman's Science Magazine

PRACTICE QUESTIONS

1. See the figures given below.

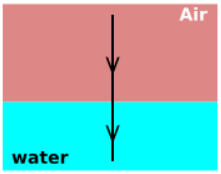


a. Find the correct figure from the three.

b. "When light incident obliquely from glass to water, angle of incidence will be greater than angle of refraction" Comment to this statement.

Ans.a. Fig.B is correct. When light travels from a rarer medium to a denser medium, refracted ray bends towards the normal.

b. This statement is incorrect. When light travels from a denser medium to a rarer medium, the refracted ray bends away from the normal, making the angle of refraction greater than the angle of incidence.



2. Light is incident from air to water as shown.

The angle of incidence is ..... (zero, 90, none of these) **Ans**. zero

3. When light passed from a medium A to B, the angle of refraction was greater than the angle of incidence.

a. Which is the medium of greater optical density.

b. Which is the medium of higher refractive index?

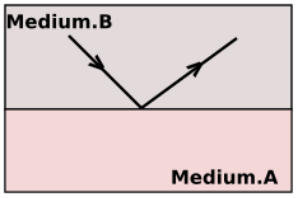
Ans.a. Medium A [Since the angle of refraction is greater than the angle of incidence, it is certain that light is travelling from a denser medium to a rarer medium]

b. Medium A. [If the optical density is greater, the refractive index will be higher]

4. Refractive index of glass is 3/2. Calculate the speed of light through glass.

Given speed of light in vacuum is $3x10^8$ m/s **Ans.**n = c/v Or v = c/n = $3x10^8/(3/2) = 2x10^8$ m/s

5. A light ray from medium B falls on another transparent medium A and is reflected back to the first medium.



a. What is called this phenomenon?

b. Identify the medium of greater optical density.

c. Which is the medium of higher refractive index?

d. In which medium does light travels with greater speed?

Ans.a.Total internal reflection. b. Medium B. c. Medium B. d. Medium A.

Layman's Science Magazine

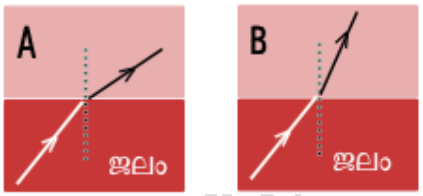
7. We can see many small prisms in the reflectors used on motor cycles. Describe the benefit of using them.

Ans. Reflectors on bikes help motorists by increasing the visibility of the bike, especially in low light conditions. When light from a vehicle or street lights hits the reflectors on a bike, it bounces back to the source due to total internal reflection at the minute prisms arranged in the reflector.

Speed of light in kerosene is lesser than that in water. Which medium has lower refractive index?

Ans. Water. Because wherever refractive index is low, speed of light will be greater there.

9. A ray of light enters from water to the media A & B at the same angle of incidence as shown.



a. Identify the phenomenon taking place here.

b. Of the media A&B, which is optically denser? Justify your answer.

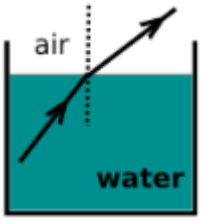
c.What is the relation between optical density of a medium and speed of light through it?

Ans.a. Refraction.

b. Medium B is optically denser than that of A. Because the refracted ray bends towards the normal in this medium, it indicates that the medium B is denser than water.

c. The speed of light decreases with an increase in optical density.

10. Analyse the given diagram.



a. Here the refracted ray deviates.....

(towards the normal/ away from the normal)

b. Name the phenomenon responsible for this kinds of deviation?

c. What happens to the speed of light when it enters from water to air?

(increases, decreases, no change)

Ans.a. away from the normal. b. Refraction. c. Increases.

11. Refractive indices of a few transparent substances are given.

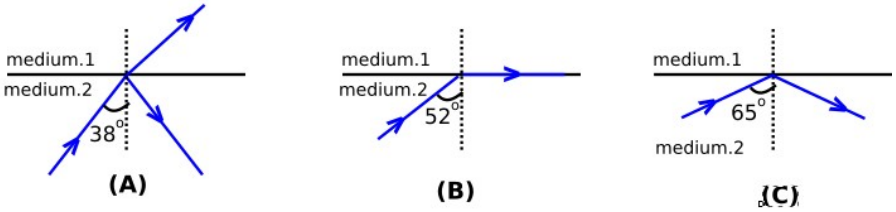
water - 1.33, flint glass - 1.62, crown glass - 1.52, sun flower oil - 1.47.

a. What is meant by refractive index?

b. Arrange the given substances in the ascending order of speed of light through them.

Ans.a. The refractive index of a medium is the ratio of the speed of light in a vacuum to the speed of light in that medium. b. Flint glass,crown glass,sunflower oil, water.

12. Which of the following figure represents the critical angle?



Ans. Fig.B

Layman's Science Magazine

13. Identify the phenomenon of light in respect of the following.

a. Twinkling of stars.

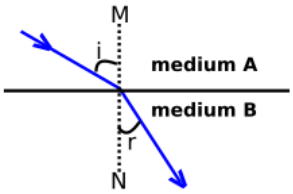
b. The bottom of the aquarium is seen at the surface of water.

c. The coin kept under a pool of water is appeared to be raised.

d. Optical fibre.

Ans.a. Refraction. b. Total internal reflection. c. Refraction d. Total internal reflection.

14. Light falls obliquely from one medium to another is shown in the figure. MN is normal at the point of incidence. Which of the given medium is the faster medium for light? Justify your answer.



Ans. The refracted ray bending towards the normal indicates that light travelled from a rarer medium to a denser medium. Thus, A is the less dense medium, meaning A is the faster medium for light.

15. Critical angle for glass is 42°. Which of the following situations involves total internal reflection?

a. When light travels from glass to air at an angle of incidence less than 42°

b. When light travels from air to glass at an angle of incidence 42°

c. When light travels from glass to air at an angle of incidence greater than 42°

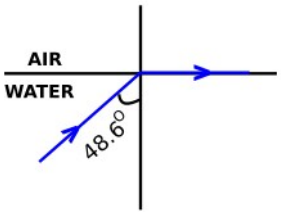
d. When light travels from air to glass at an angle of incidence greater than 42°

Ans.c.When light travels from glass to air at the angle of incidence greater than 42°

16. Critical angle of water with air is shown in the figure.

a. Define critical angle.

b. What change in the path of light will be observed if the angle of incidence is greater than critical angle?



Ans.a.When light travels from a medium of higher optical density to a medium of lower optical density, the angle of incidence at which the angle of refraction 90° is called Critical angle.

b. The ray will be reflected back to the same medium./Undergoes total internal reflection.

17. Refractive indices of some materials are given.

Glass: 1.52 Glycerine:1.47 Sunflower oil:1.47 Water:1.33 Flint glass:1.62

a. Identify a pair of materials in which light does not undergo refraction when passing from one to the other.

b.Find out the medium from them through which light passes with greatest speed?

c. Find out the medium having least optical density?

Ans.a. Glycerine and Sunflower oil (Because their is no difference in their refractive indices)

b. Water. c. Water.

Layman's Science Magazine

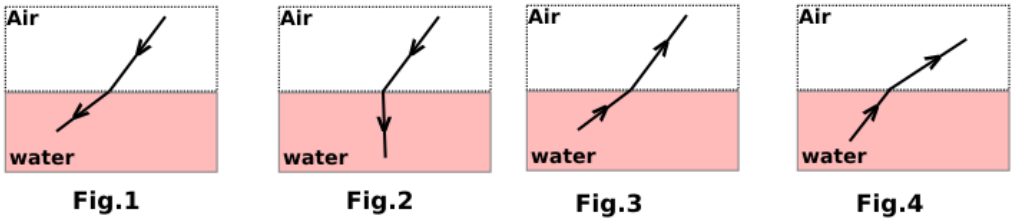
MULTIPLE CHOICE QUESTIONS

1.Refractive index of a medium is 1.2. Which of the following is the speed of light through that medium.

a. $2x10^8$ m/s b. $2.2x10^8$ m/s c. $2.5x10^8$ m/s d. $2.6x10^8$ m/s **Ans**. c. $2.5x10^8$ m/s

Explanation: v = c/n = 3x108/1.2 = $2.5x10^8$ m/s

2. Find the correct figure from the following.



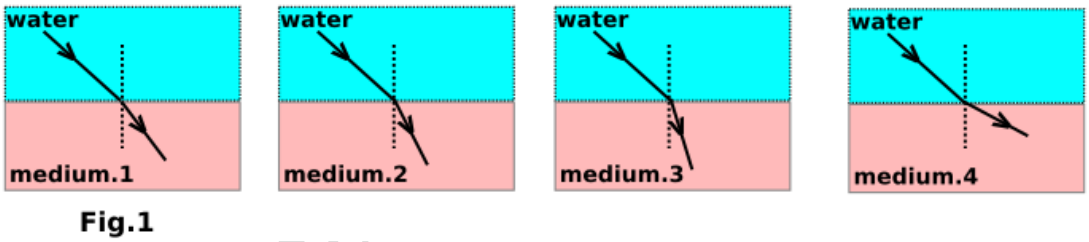
a. Fig.1 b. Fig.2 c. Fig.3 d. Fig.4 **Ans**. d. Fig.4

3. When light is incident from water to air at an angle of incidence of 40°, which of the following is likely to be the angle of refraction?

a. 59° b. 28° c. 25° d. 24° **Ans**. a. 59°

Explanation: When light travels from a denser medium to a rarer medium, the angle of refraction will be greater than the angle of incidence.

4. Light is incident from water to four different mediums at the same angle of incidence as shown.



Of these mediums, which has highest refractive index?

a. medium.1 b. medium.2 c. medium.3 d. medium.4 **Ans**. c. Medium.3

Explanation: Medium 4 cannot be the answer because the refracted ray bends away from the normal, indicating that Medium 4 is less dense than water. Among the remaining three mediums, the refracted ray has the greatest deviation (smallest angle of refraction) in Medium 3, which suggests it has the highest refractive index or optical density

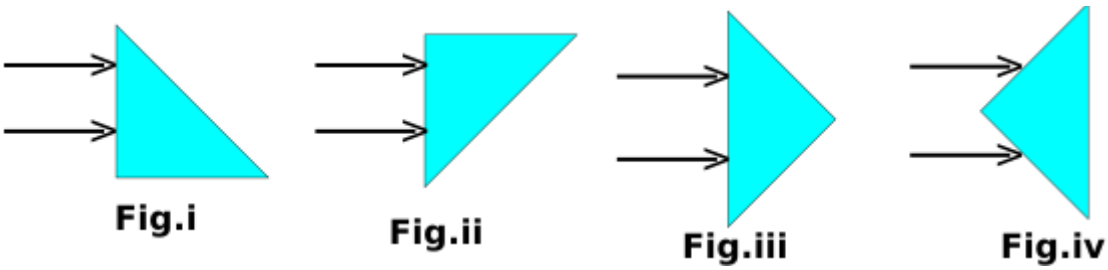
5. Twinkling of stars is due to .....

a. Reflection b. Refraction c.total internal reflection d. none of these

Ans. b. Refraction

Laymen Science Magazine

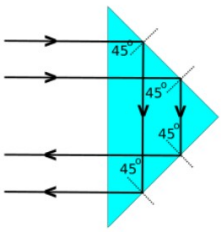
6.For working a right angled prism as a reflector on vehicles, how the light is to be incident on the prism.



a. as in Fig.i b. as in Fig.ii c. as in Fig.iii d. as in Fig.iv

Ans. as in Fig.iii

Explanation:



7. The critical angle of a pair of media is 38°. At which of the following angles of incidence is total internal reflection likely to occur?

a. 32° b.36° c.38° d.42° **Ans**. d.42°

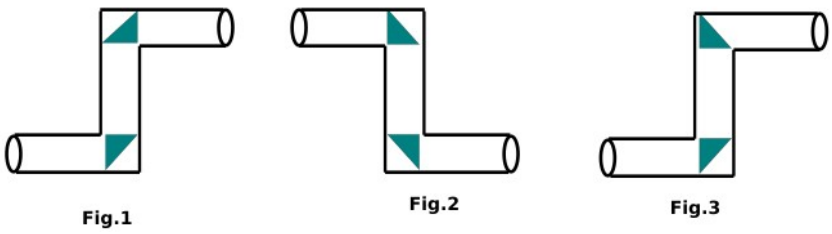
Explanation:Total internal reflection occurs only when the angle of incidence exceeds the critical angle.

8. Light undergoes total internal reflection when the angle of incidence is 40°. Which of the following is likely to be its critical angle.

a.35° b.40° c.42° d. 48.6° **Ans**. a.35°

Explanation:Here total internal reflection occurs at 40°, indicating that the critical angle is less than 40°

9. Which of the following cannot work as periscope?



a.Fig.1 b.Fig.2 c.Fig.3 d. Fig.1& 3 **Ans**. c.Fig.3

10.The critical angle for air-glass media is 42°. If a light ray from air is incident at 42° on the surface of water, what is the likely angle of refraction?

a. 90° b. 50° a. 42° a. 25° **Ans**.a. 25°

Explanation: When light travels from a rarer medium to a denser medium (air to water), the angle of refraction is less than the angle of incidence.

Layman's Science Magazine